

House Price Prediction using Machine Learning

Project Overview

This project predicts residential house prices using machine learning techniques. Various regression algorithms are trained and evaluated, with the best-performing model deployed through a professional Flask web application.

Dataset Overview

The dataset contains detailed information about residential properties, including physical characteristics, location, quality, condition, and sale information.

Each row represents **one house**, while each column represents a specific property attribute.

Attribute	Value
Dataset Name	House Prices – Advanced Regression Techniques
Dataset Type	Structured Tabular Data
Domain	Real Estate
Prediction Type	Regression
Target Variable	SalePrice
Number of Records	1,460
Total Columns	81
File Used	train.csv

Phase 1 -Data Understanding

The **House Prices – Advanced Regression Techniques** dataset contains **1,460 records** and **81 columns**, comprising **38 numerical features**, **43 categorical features**, and **1 target variable (Sale Price)**. The dataset includes property-related information such as location, size, construction quality, condition, and sale details, which are used to predict the final selling price of a house.

Missing Values

The dataset contains missing values in features such as **LotFrontage**, **Alley**, **MasVnrType**, **MasVnrArea**, **Basement**, **Garage**, **PoolQC**, **Fence**, and **MiscFeature**. These missing values were handled using appropriate imputation techniques based on the feature type to ensure data completeness.

Duplicate Records

The dataset was checked for duplicate records, and no duplicate entries were found. Therefore, no rows were removed before further preprocessing and analysis.

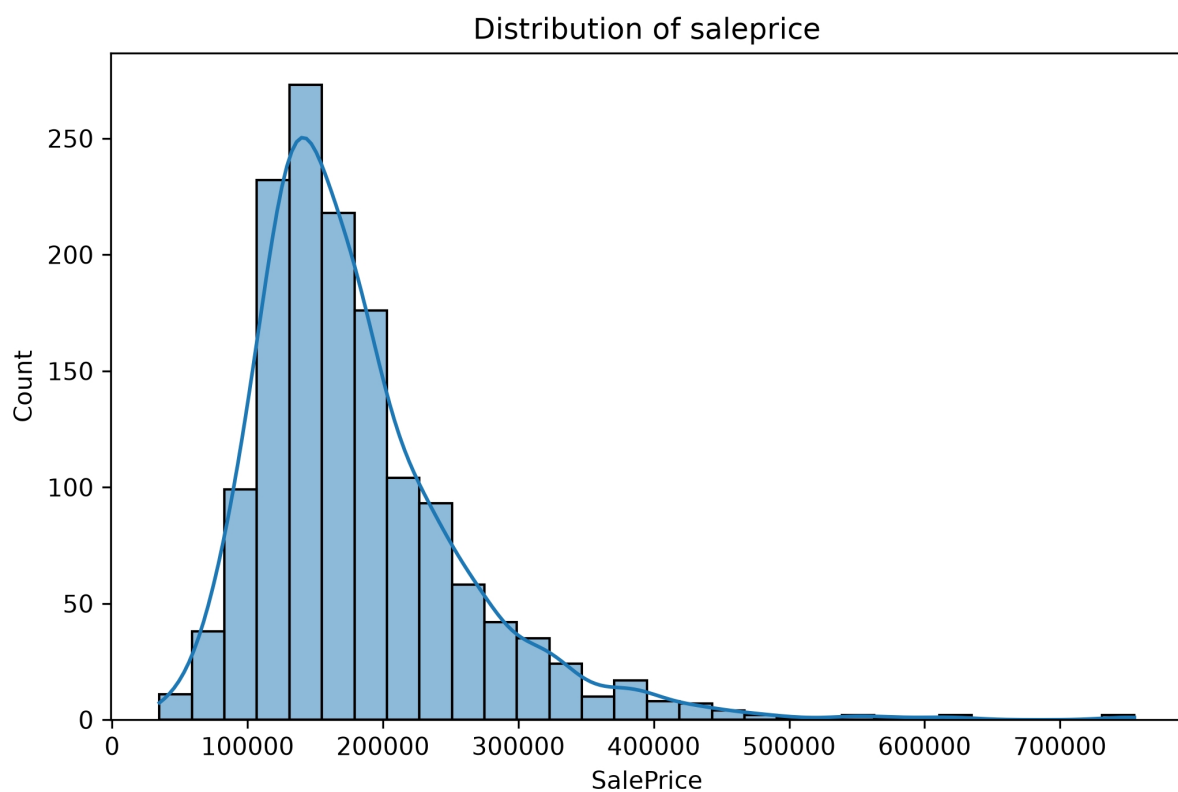


Figure 1: Saleprice Distribution

Phase 2 – Data Preprocessing

- Handled missing values
- Removed duplicates
- Treated outliers using IQR
- Encoded categorical variables
- Applied Log Transformation to skewed features

- Applied Standard Scaling to numerical features
- Performed Feature Engineering

Feature Engineering

- OverallQual
- GrLivArea
- GarageYrBlt
- GarageCars
- TotalSF
- BsmtFinSF1
- LotArea

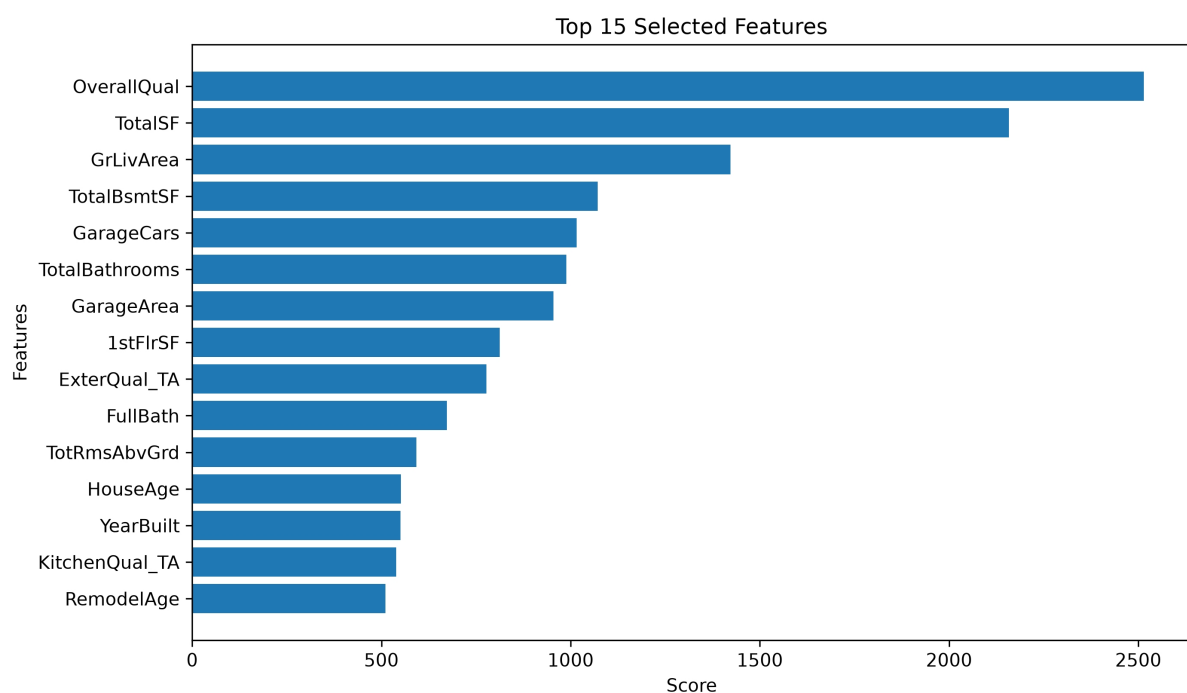


Figure 2: Top 15 Selected features

Phase 3 – Exploratory Data Analysis(EDA)

- Feature vs Target Analysis
- Correalation Analysis
- outlier analysis
- Feature Importance Exploration
- Relation Between variables

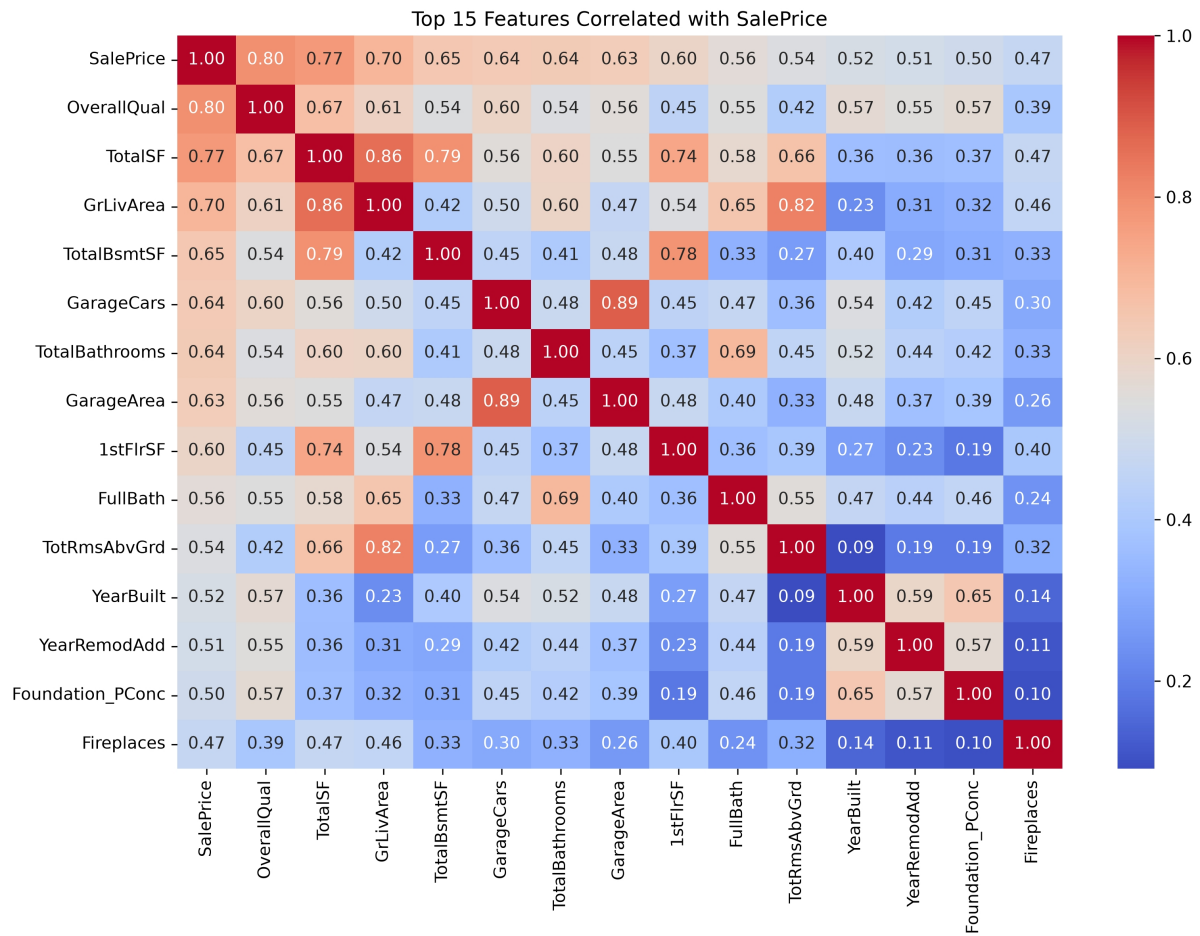


Figure 3: Correlation Heatmap

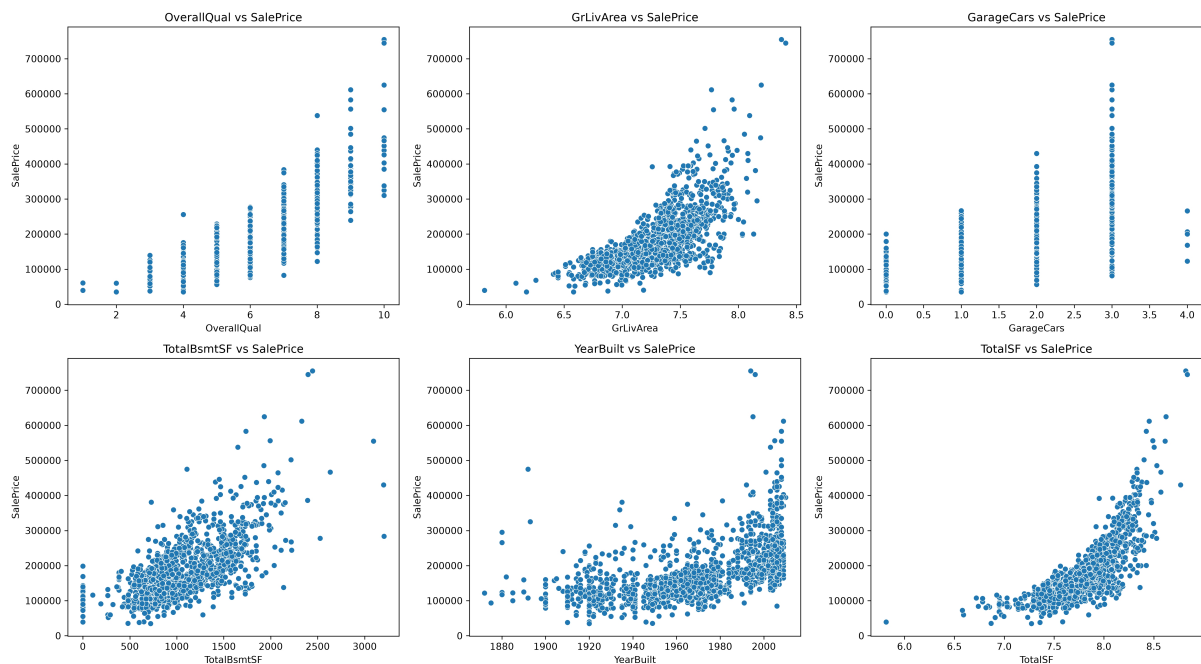


Figure 4: Correlation Heatmap

Phase 4 - Model Development

- Train Test Split
- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

Model Comparison

S.No	Model	MAE	MSE	RMSE	R ² Score
1	Linear Regression	0.22534.897952	3.015176e+09	54910.618382	0.454141
2	Decision Tree	26273.643836	1.473032e+09	38380.092897	0.733327
3	Random Forest	16595.651918	6.276960e+08	25053.862473	0.886364
4	Gradient Boosting	15228.036023	4.677599e+08	21627.756577	0.915318
5	XGBoost	16524.613281	5.605974e+08	23676.938991	0.898511
6	LightGBM	15909.718895	5.848068e+08	24182.779722	0.894128
7	CatBoost	13834.611536	3.772849e+08	19423.822517	0.931697

Outcome

CatBoost achieved the best performance.

A CatBoost Regressor was trained using the training dataset. CatBoost is an advanced gradient boosting algorithm that provides strong predictive performance and reduces overfitting. The model was evaluated using MAE, MSE, RMSE, and R² Score.

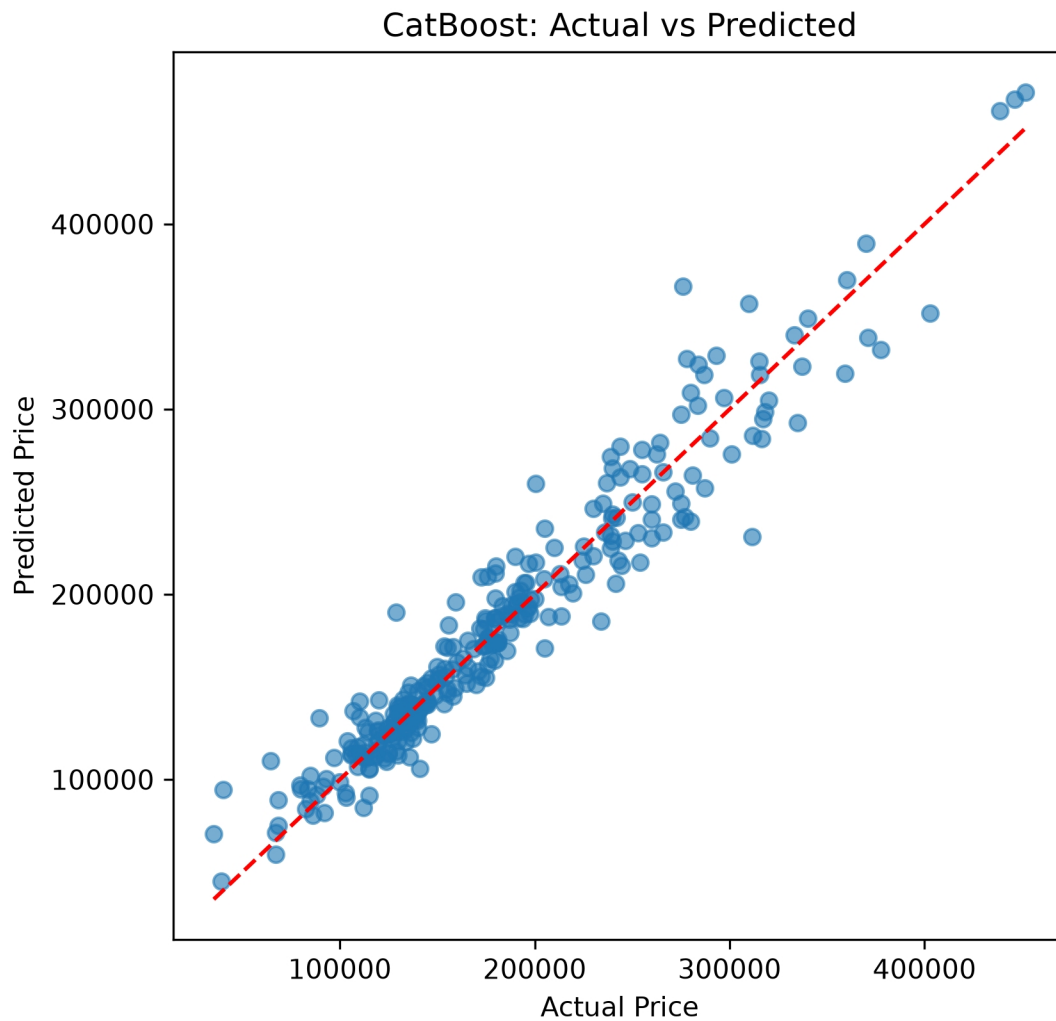


Figure 5: Actual VS Predicted

Phase 5 – Model Improvement & Optimization

Multicollinearity(VIF)

Variance Inflation Factor (VIF) analysis was performed to detect multicollinearity among the features. Although a few variables showed high or infinite VIF values due to strong correlations, they were retained because they contain valuable predictive information and contribute to the overall model performance

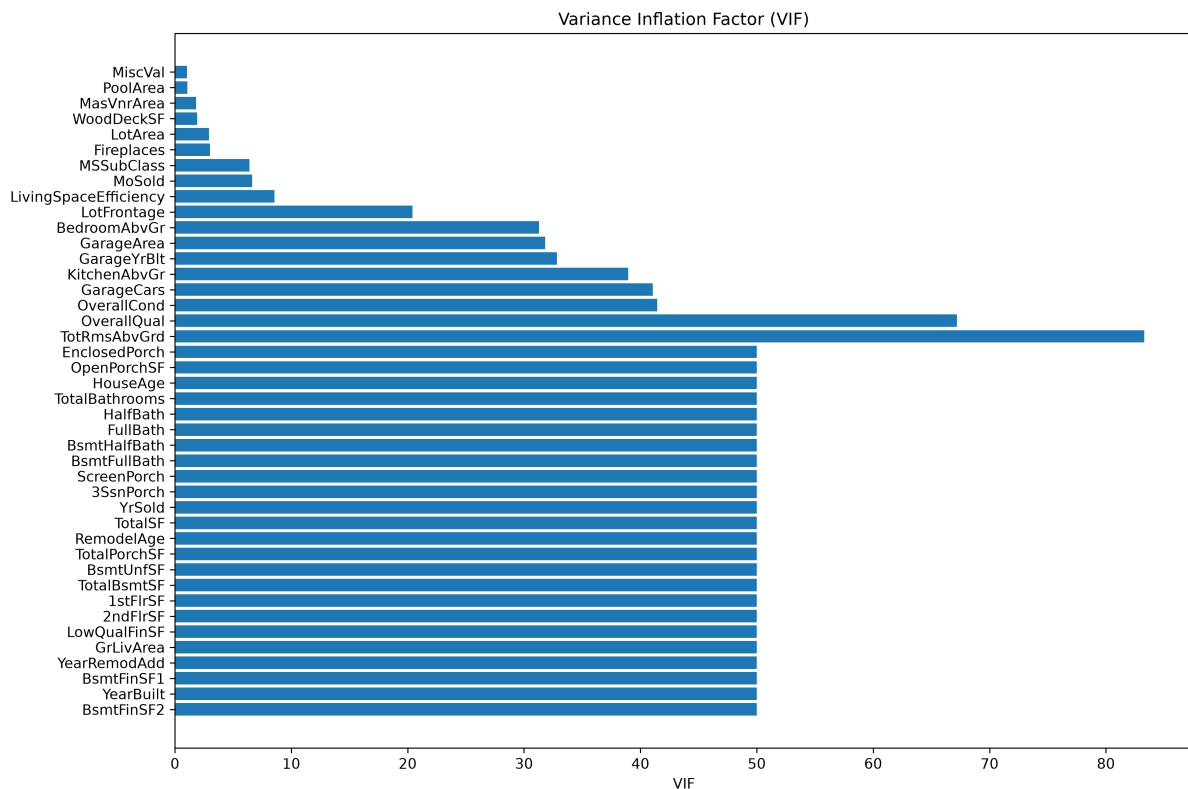


Figure 6:Variance infaltion factor(VIF)

Cross Validation

A 5-fold cross-validation was performed on the **Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, XGBoost Regressor, LightGBM Regressor, and CatBoost Regressor** models to evaluate their performance and generalization ability. The average **R² score** and **standard deviation** were calculated for each model to ensure a reliable and consistent performance evaluation.

Validation Table

No	Model	Mean R ²	Std Dev
1	Linear Regression	0.902705	0.011296
2	Decision Tree	0.743312	0.023186
3	Random Forest	0.876992	0.008083
4	Gradient Boosting	0.902667	0.005530
5	XGBoost	0.885854	0.007263
6	LightGBM	0.899311	0.005988
7	CatBoost	0.914009	0.004822

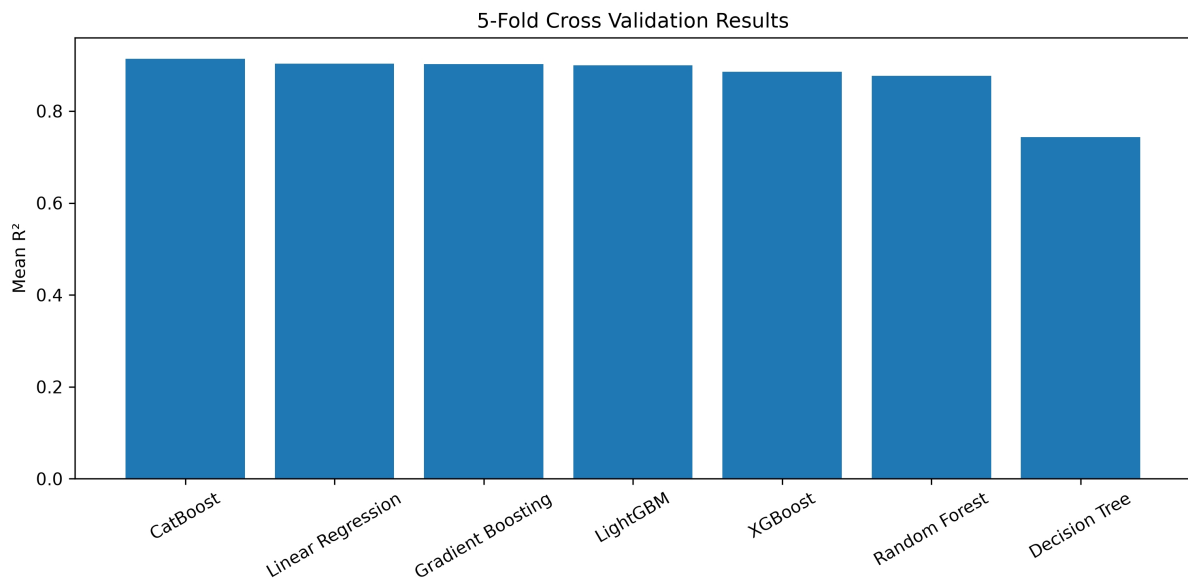


Figure 7: Cross validation

Hyperparameter Tuning

Hyperparameter tuning was performed to improve the performance of the regression models. **Linear Regression** was trained using its default configuration, as it does not have significant hyperparameters to optimize. The **Decision Tree Regressor**, **Random Forest Regressor**, **Gradient Boosting Regressor**, **XGBoost Regressor**, **LightGBM Regressor**, and **CatBoost Regressor** were optimized using **RandomizedSearchCV** with **5-fold cross-validation**. The **R^2 score** was used as the evaluation metric to identify the optimal hyperparameter combinations and improve the models' predictive performance.

Hyperparameter Search Space for Regression Models :

Model	Hyperparameters Tuned
Linear Regression	Default parameters (no hyperparameter tuning)
Decision Tree Regressor	max_depth, min_samples_split, min_samples_leaf
Random Forest Regressor	n_estimators, max_depth, min_samples_split, min_samples_leaf
Gradient Boosting Regressor	n_estimators, learning_rate, max_depth
XGBoost Regressor	n_estimators, learning_rate, max_depth
LightGBM Regressor	n_estimators, learning_rate, max_depth
CatBoost Regressor	iterations, learning_rate, depth

No	Model	MAE	MSE	RMSE	R2
1	Decision Tree (Tuned)	23058.125753	9.746211e+08	31218.921788	0.823557
2	Random Forest (Tuned)	16336.485458	6.067234e+08	24631.756382	0.890161
3	Gradient Boosting (Tuned)	15037.077031	4.668676e+08	21607.119068	0.915480
4	XGBoost (Tuned)	14809.859375	4.396696e+08	20968.300646	0.920403
5	LightGBM (Tuned)	15441.468325	4.997429e+08	22354.929249	0.909528
6	CatBoost (Tuned)	14311.229939	3.977711e+08	19944.198843	0.927989

Among the tuned regression models, **CatBoost Regressor** achieved the best performance with the **highest R² score (0.9280)** and the **lowest MAE, MSE, and RMSE**. Therefore, it was selected as the final model for deployment due to its superior prediction accuracy and overall performance.

Phase 6 - Model Evaluation & Comparison¶

The optimized regression models were evaluated using MAE, MSE, RMSE, R² Score, Adjusted R², and 5-fold Cross-Validation Score. These metrics were used to compare the predictive accuracy and generalization performance of each model on the test dataset.

Model Evaluation Results

Model	MAE	MSE	RMSE	R ² Score	Adjusted R ²	CV Score
CatBoost	14,311.23	3.98×10^8	19,944.20	0.9280	0.8983	0.9098
XGBoost	14,809.86	4.40×10^8	20,968.30	0.9204	0.8876	0.9057
Gradient Boosting	15,037.08	4.67×10^8	21,607.12	0.9155	0.8806	0.9023
LightGBM	15,441.47	5.00×10^8	22,354.93	0.9095	0.8722	0.9071
Random Forest	16,336.49	6.07×10^8	24,631.76	0.8902	0.8448	0.8783
Decision Tree	23,058.13	9.75×10^8	31,218.92	0.8236	0.7508	0.7793

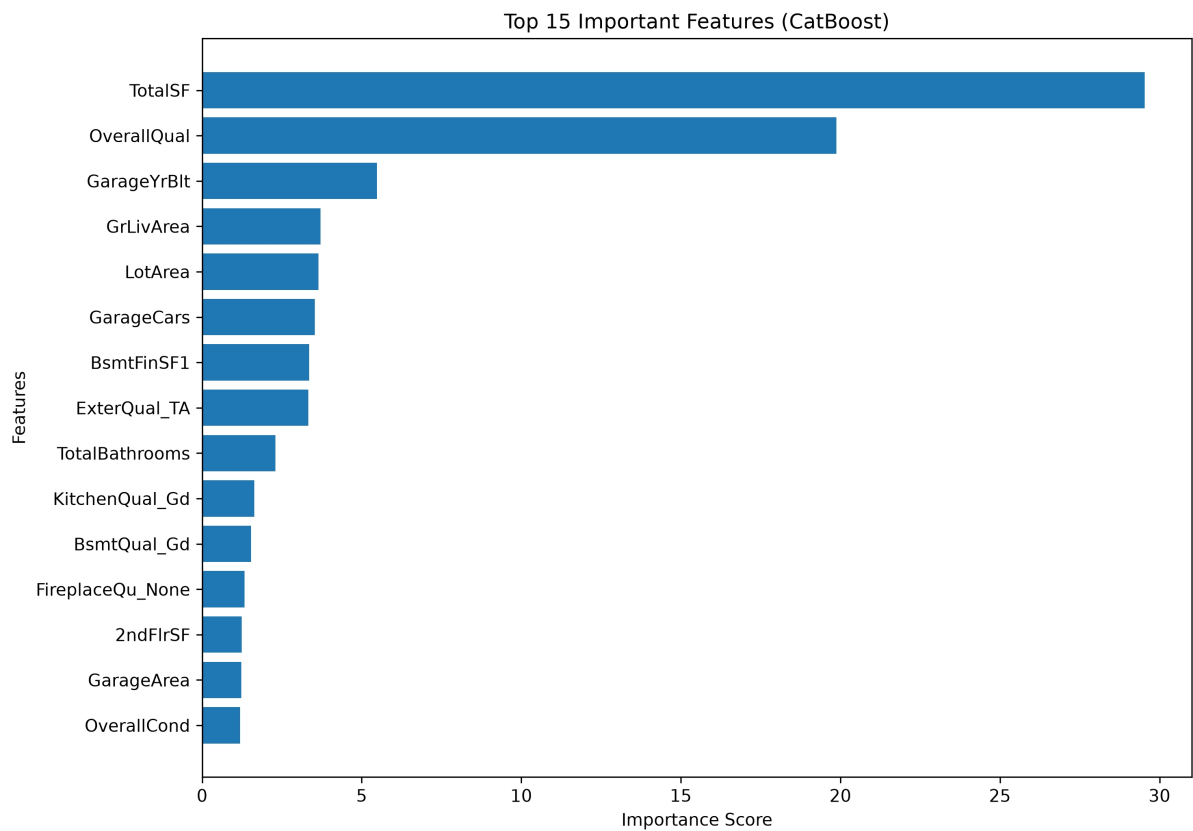


Figure 8: Catboost Important Features

Best Model Selection

Among all evaluated regression models, CatBoost achieved the best overall performance. It produced the highest R^2 Score (0.927989) and Adjusted R^2 (0.898275), while also achieving the lowest MAE, MSE, and RMSE values. The 5-fold Cross Validation Score of 0.909774 demonstrates that the model generalizes well to unseen data. Therefore, CatBoost was selected as the final model for house price prediction.

Phase 7 – Model Interpretation

Business Objective

Develop and compare multiple regression models to accurately predict house prices. This identifies the best-performing model through preprocessing, optimization, and evaluation to support reliable property valuation.

Model Assumptions

- Historical data represents the housing market.
- Selected features influence house prices.
- Missing values and outliers are handled.
- Training and testing data have similar distributions.
- The model generalizes well to unseen data.

Feature Importance

Feature importance was analyzed using the **CatBoost** model. **TotalSF**, **OverallQual**, **HouseAge**, **TotalBathrooms**, and **GrLivArea** were identified as the most influential features for predicting house prices.

Regression Model Comparison

All models were evaluated using **MAE**, **MSE**, **RMSE**, **R^2 Score**, **Adjusted R^2** , and **Cross-Validation Score**. **CatBoost** achieved the best overall performance and was selected as the final model.

Advantages and Limitations

Model	Advantage	Limitation
Linear Regression	Simple and fast	Poor on non-linear data
Decision Tree	Easy to interpret	Overfitting
Random Forest	Robust and accurate	Higher training time
Gradient Boosting	High accuracy	Slower training
XGBoost	Excellent performance	Complex tuning
LightGBM	Fast and efficient	Sensitive to parameters
CatBoost	Highest accuracy	Higher computational cost

Final Model Selection

CatBoost was selected as the final model because it achieved the **highest R^2 Score (0.927989)** and the **lowest prediction errors**, making it the most accurate model.

Business Recommendations

- Use **CatBoost** for house price prediction.
- Retrain the model with new market data.
- Focus on important features such as **TotalSF** and **OverallQual**.
- Deploy the model as a **Flask web application** for real-time predictions.

Phase 8 – Flask Web Application

Overview

The Flask web application provides an interactive platform for predicting house prices, comparing regression models, and exploring analytical insights using the trained CatBoost model.

Dashboard

- Dataset Overview
- Model Comparison Summary
- Performance Metrics

Prediction Module

- Property Details Input
- House Price Prediction
- Prediction Result
- Important Feature Contribution
- Shap explanation

Model Comparison

- Regression Model Comparison
- Evaluation Metrics
- Best Model Highlight

Analytics Dashboard

- Interactive EDA
- Correlation Heatmap
- Feature Distributions
- Feature Importance
- Residual Analysis

Reports

- Prediction Report
- Model Evaluation Report
- Model Comparison Report

- EDA report
- Data Preprocessing Report

Technologies Used

- Python
- Flask
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- LightGBM
- CatBoost
- Matplotlib
- Seaborn
- HTML5
- CSS3
- Bootstrap 5
- Git
- GitHub

Application Workflow

User Input



Data Preprocessing



CatBoost Model



House Price Prediction



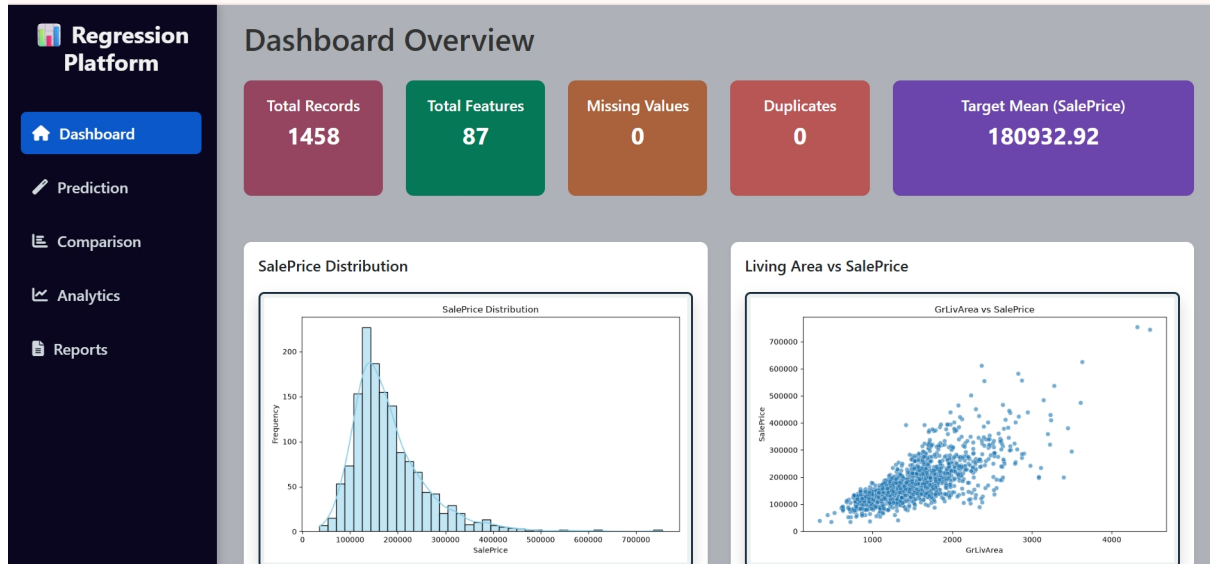
SHAP Explainability



Display Prediction, Feature Contributions

Screenshots

Home dashboard



Prediction Page

The prediction page allows users to input various house features to predict the sale price. The form includes fields for Overall Quality, Living Area (sq.ft), Total Square Feet, Garage Cars, Garage Year Built, Basement Finished Area (sq.ft), and Lot Area. A 'Predict Price' button is located at the bottom of the form.

Regression Platform

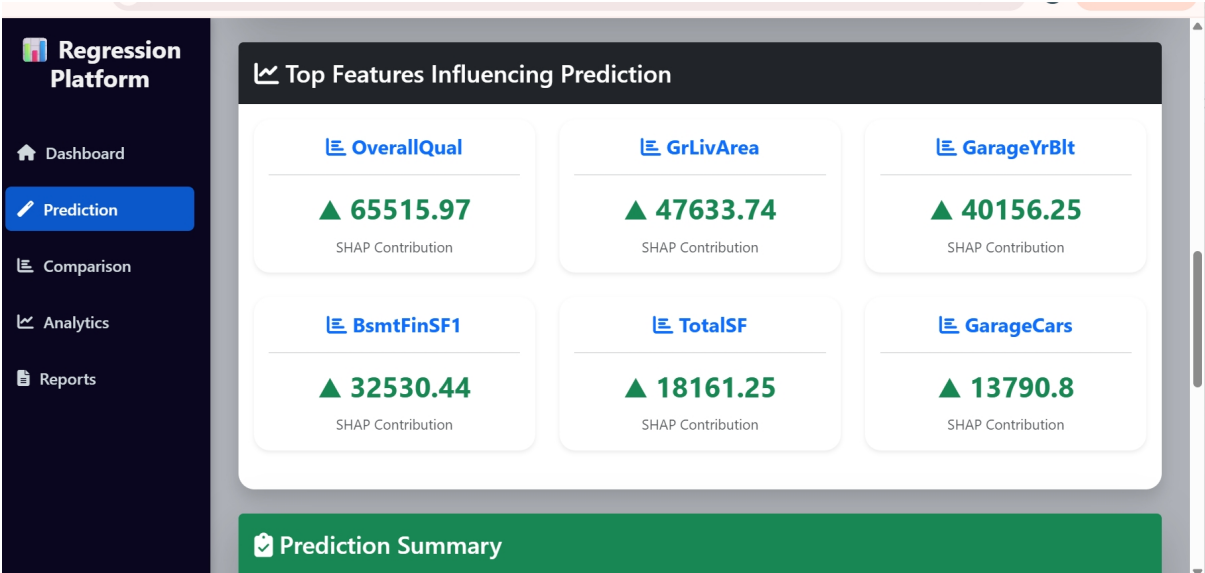
- Dashboard
- Prediction
- Comparison
- Analytics
- Reports

House Price Prediction

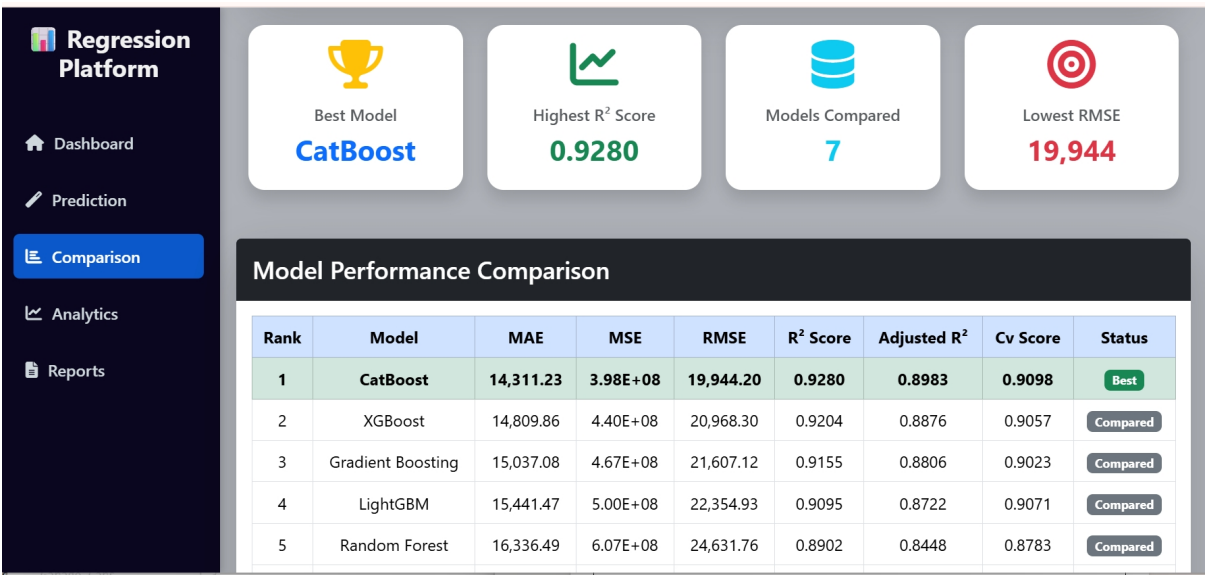
Overall Quality	Living Area (sq.ft)
9	2800
Total Square Feet	Garage Cars
3600	3
Garage Year Built	Basement Finished Area (sq.ft)
2010	1400
Lot Area	
9500	

Predict Price

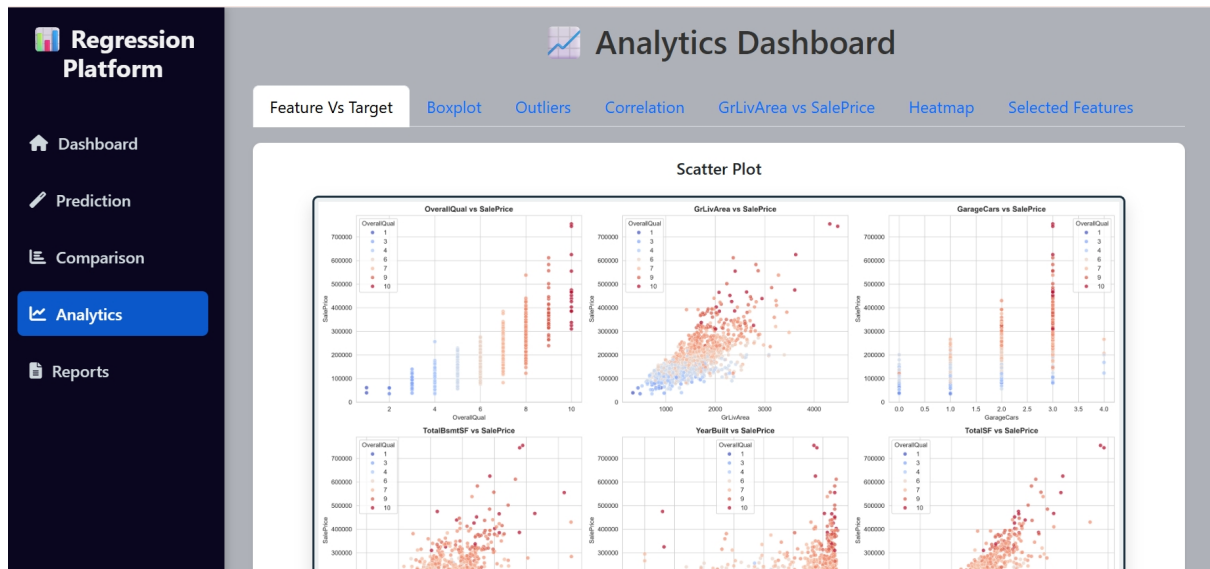
SHAP Explainability



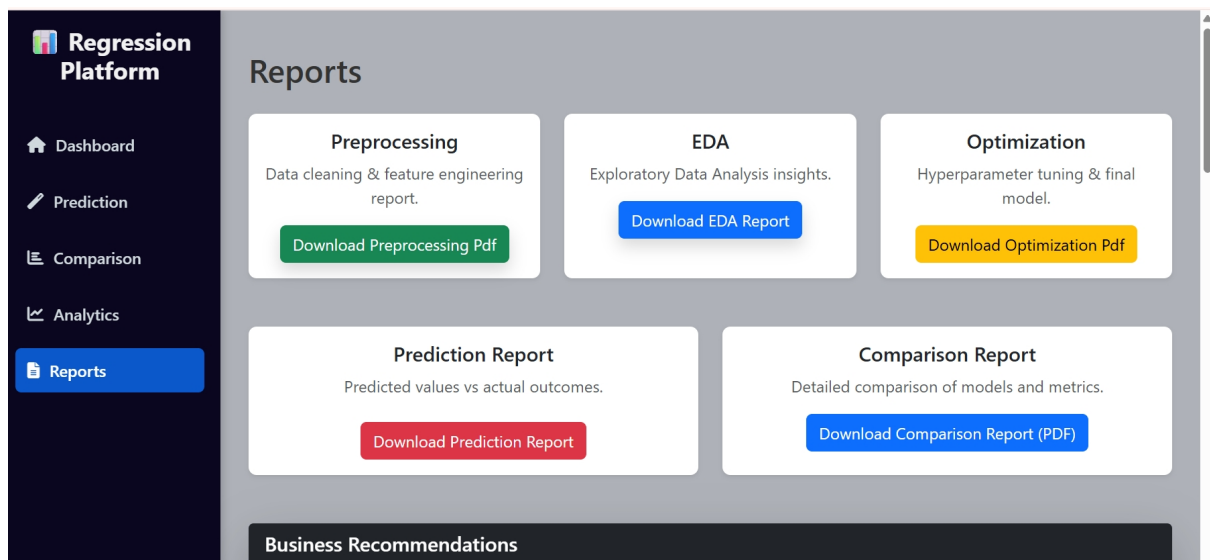
Model Comparison



Analytics Dashboard



Reports



Conclusion

A complete machine learning solution was developed to accurately predict house prices using multiple regression models. After preprocessing, feature engineering, hyperparameter tuning, and performance evaluation, the **CatBoost Regressor** was selected as the final model due to its superior accuracy. The model was deployed through a **Flask web application**, providing real-time house price prediction, SHAP-based feature explanations, model comparison, analytics, and report generation in an interactive and user-friendly